

Samuel Roth

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Project 2 Aerodynamics

Abstract

This project uses the source panel method to model inviscid, incompressible, and irrotational flow over a NACA 0012 airfoil. The airfoil's surface was divided into source panels and solved at each point. Surface velocity, pressure coefficient distributions, and flow field visualizations were computed for both $\alpha = 0^\circ$ and $\alpha = 10^\circ$ and compared with experimental data. At zero angle of attack, the method was reasonably accurate and showed agreement with experimental results, but at a nonzero angle of attack, the method did capture asymmetry but failed to accurately predict the pressure distribution, particularly at the leading edge. Flow visualizations also showed unrealistic behavior near the trailing edge. Mass imbalance error was studied which showed an improvement in accuracy with increasing panel count. The method was accurate for general flow but is limited by a lack of circulation and viscous effects, which reduces its accuracy for lifting flows.

Technical Approach

The source panel method is a technique used to approximate inviscid, incompressible, irrotational flow around a body by replacing the body surface with a distribution of single sources. In this project, the airfoil surface is discretized into a finite number of panels and then assigned a source strength λ_j . Once the source strengths are known, pressure coefficient distribution can be computed and compared to experimental data as well as analyzed for its accuracy.

The main purpose of the source panel method is to model the flow field around a geometry while enforcing the body's boundary. For a closed body, this method shows how the flow is deflected by the body's surface and can provide predictions of surface pressure trends, especially at negligible angles of attack and without viscous effects.

This method, however, has its limitations. Since it is based on potential flow, it does not include viscosity, flow separation, and wake. Only accounting for the source also does not have any circulation and therefore cannot properly predict lift for a lifting airfoil unless a vortex distribution or Kutta condition is added. As a result, the method can approximate pressure distribution and streamlines, but it is not as accurate near the trailing edge, with various angles of attack, and in any situation where viscous or rotational effects occur. Comparison to experimental data will therefore show discrepancy.

The first step taken in order to implement the source panel method was to define the geometry of the airfoil being used. This was created using the NACA 00xx thickness equation where the "xx" is replaced by the percentage of thickness to chord. In this instance, $t = 0.12$ which corresponded to the design parameter of a NACA 0012-type airfoil. The thickness distribution used in the code is

$$y_t = \pm 5 t (0.2969 \sqrt{x} - 0.1260 x - 0.3516 x^2 + 0.2843 x^3 - 0.1015 x^4)$$

Because there is no camber on the airfoil as defined by the zeros in NACA 0012, we are able to use this equation for both the top and bottom of the airfoil by using the positive version of the equation for the top and negative for the bottom, where:

$$\begin{aligned} y_{upper} &= + y_t(x) \\ y_{lower} &= - y_t(x) \end{aligned}$$

The full airfoil surface was divided into N_{Panels} by distributing $\frac{N}{2} + 1$ nodes along each surface from leading edge to trailing edge and trailing edge back to leading edge which created a closed contour. Each panel j , is defined by its two endpoints: (X_j, Y_j) and (X_{j+1}, Y_{j+1}) .

The length of the panel can be defined by:

$$S_j = \sqrt{(X_{j+1} - X_j)^2 + (Y_{j+1} - Y_j)^2}$$

and the panel inclination angle relative to the x-axis is

$$\phi_j = \tan^{-1}\left(\frac{Y_{j+1} - Y_j}{X_{j+1} - X_j}\right)$$

The panel midpoint (x_j, y_j) is the control point where the boundary conditions are enforced.

$$x_j = \frac{X_j + X_{j+1}}{2}, y_j = \frac{Y_j + Y_{j+1}}{2}$$

The outward normal direction in degrees is taken as

$$\beta_j = \phi_j + 90^\circ$$

The governing condition for the source panel method is the no-penetration condition at each point, which requires the normal component of total velocity to be zero at all panels:

$$v_n = \vec{V}_\infty \cdot \hat{n}_i + \frac{\lambda_i}{2} + \sum_{j=1; i \neq j}^n \frac{\lambda_j}{2\pi} \int_j \frac{\partial}{\partial n} \ln r_{ij} ds = 0$$

where $\vec{V}_\infty$ is the freestream velocity, $\hat{n}_i$ is the outward unit normal at point i , λ_j is the source strength on panel j , and the integral represents the normal velocity at point i by source panel j .

For a constant-strength source panel, this becomes

$$\sum_{j=1; i \neq j}^n I_{ij} \lambda_j = -2\pi \cdot \vec{V}_\infty \cos(\beta_i - \alpha)$$

where α is the angle of attack, which represents the angle between the chord and oncoming airflow, and λ_j is the unknown that is being solved for from the right hand side, which is determined by freestream normal velocity. This is solved in the code through

$$\lambda = I^{-1}b, \text{ where } b = -2\pi * \vec{V}_\infty \cos(\beta_i - \alpha) * \hat{n}_i$$

To find the influence of each source panel on each point, the code first solves for these values

$$\begin{aligned} A_{ij} &= -[(x_i - X_j) \cos \phi_j + (y_i - Y_j) \sin \phi_j] \\ B_{ij} &= (x_i - X_j)^2 + (y_i - Y_j)^2 \\ C_{ij} &= \sin(\phi_i - \phi_j) \\ D_{ij} &= (y_i - Y_j) \cos \phi_i - (x_i - X_j) \sin \phi_j \\ E_{ij} &= (x_i - X_j) \sin \phi_j - (y_i - Y_j) \cos \phi_i \end{aligned}$$

After solving for these values, we can bring S_j to solve for the the normal influence integral, with the equation:

$$I_{ij} = \frac{C_{ij}}{2} \ln\left(\frac{S_j^2 + 2A_{ij}S_j + B_{ij}}{B_{ij}}\right) + \frac{D_{ij} - A_{ij}C_{ij}}{E_{ij}} \left[\tan^{-1}\left(\frac{S_j + A_{ij}}{E_{ij}}\right) - \tan^{-1}\left(\frac{A_{ij}}{E_{ij}}\right)\right]$$

We manually set the diagonal of the matrix to π so that each panel can account for its own influence on itself, without causing a mathematical issue:

$$I_{ii} = \pi$$

After solving for the source strengths, the tangential velocity on the airfoil surface was found. This first requires solving for the integral of the j panel similar from before, resulting in the equation:

$$J_{ij} = \frac{D_{ij} - A_{ij}C_{ij}}{2E_{ij}} \ln\left(\frac{S_j^2 + 2A_{ij}S_j + B_{ij}}{B_{ij}}\right) - C_{ij} \left[\tan^{-1}\left(\frac{S_j + A_{ij}}{E_{ij}}\right) - \tan^{-1}\left(\frac{A_{ij}}{E_{ij}}\right)\right]$$

with the diagonal term set to

$$J_{ii} = 0$$

The tangential velocity from by all panels at point i is then

$$V_{s,i} = \sum_{j=1}^n \frac{J_j \lambda_j}{2\pi}$$

The total local tangential velocity is obtained by adding the tangential component of the freestream:

$$V_i = V_{\infty} \sin(\beta_i - \alpha) + V_{s,i}$$

Finally, the pressure coefficient can be derived from Bernoulli's equation for incompressible potential flow:

$$C_{p,i} = 1 - \left(\frac{V_i}{V_{\infty}}\right)^2$$

This distribution was then compared against the experimental pressure coefficient data at $\alpha = 0^\circ$ and $\alpha = 10^\circ$.

The flow field is then visualized by initially finding all velocities on a Cartesian grid surrounding the airfoil with the freestream contribution initialized as

$$u = V_{\infty} \cos \alpha, \quad v = V_{\infty} \sin \alpha$$

Each panel was then broken into a large number of small source elements, and the velocity from each source element was added to the grid. For a source element of strength λ_s at (x_s, y_s) , the velocity components at a point (x_p, y_p) are

$$u_s = \frac{\lambda_s}{2\pi} \frac{x_p - x_s}{(x_p - x_s)^2 + (y_p - y_s)^2}$$

$$v_s = \frac{\lambda_s}{2\pi} \frac{y_p - y_s}{(x_p - x_s)^2 + (y_p - y_s)^2}$$

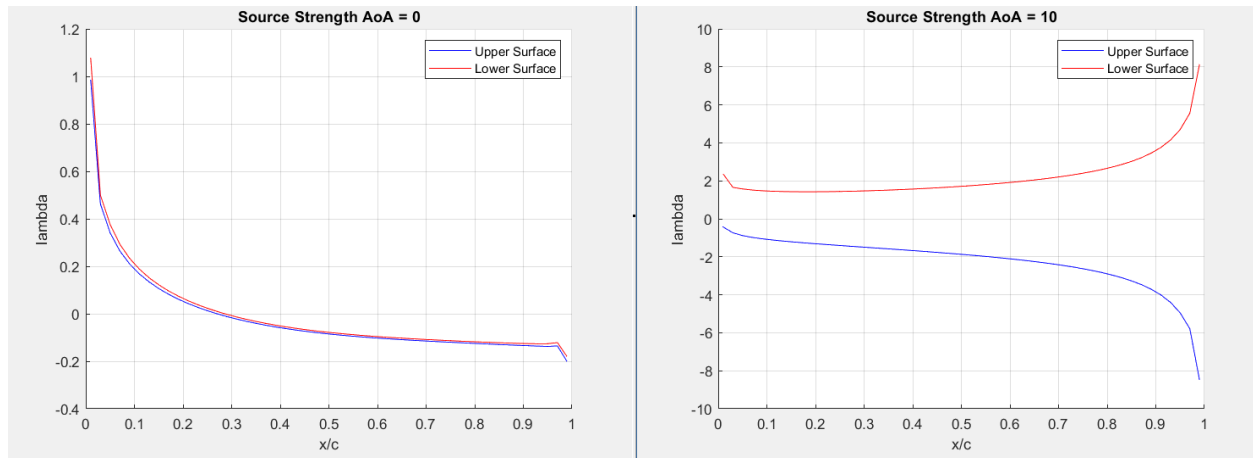
The total velocity field was then used to plot streamlines and contours of velocity magnitude.

Since the airfoil is a closed body, the sum of all the source and sink strengths must equal to zero, or else the body would be adding or absorbing mass from the flow, which is impossible for this case. This allows us to check the solution by calculating:

$$\sum_{j=1}^n \lambda_j S_j = 0$$

For an ideal solution, this should approach zero as the number of panels increases.

Results



Figures 1-2: Source strength distribution for a NACA 0012 airfoil at angles of attack $\alpha = 0^\circ$ and $\alpha = 10^\circ$ using 100 panels. Symmetry seen at zero angle of attack, while increasing angle of attack produces asymmetry and unrealistic results.

The plotted source strength distributions represent how the source panel method represents flow around a symmetric airfoil. At $\alpha = 0^\circ$, the upper and lower surface distributions overlap, which accurately reflects the symmetry and lack of camber in the NACA 0012 airfoil. The largest source strengths occur near the leading edge, where flow experiences the largest deceleration and as the flow moves towards the trailing edge, the source strength decreases smoothly and approaches small negative values. This is consistent with what is expected for the flow and shows that this simulation is accurate.

At $\alpha = 10^\circ$, there is asymmetry, and the source strength distributions on the upper and lower surfaces are significantly different. The upper surface has negative values of λ , while the lower surface shows positive values that increase towards the trailing edge. This shows the change in the flow response to the airfoil, however, the results also represent the flaws of the source panel method. Due to the fact that circulation is not included and the Kutta condition is absent, the method cannot accurately represent lifting flow. The behavior is unrealistic near the trailing edge as the source strength becomes large.

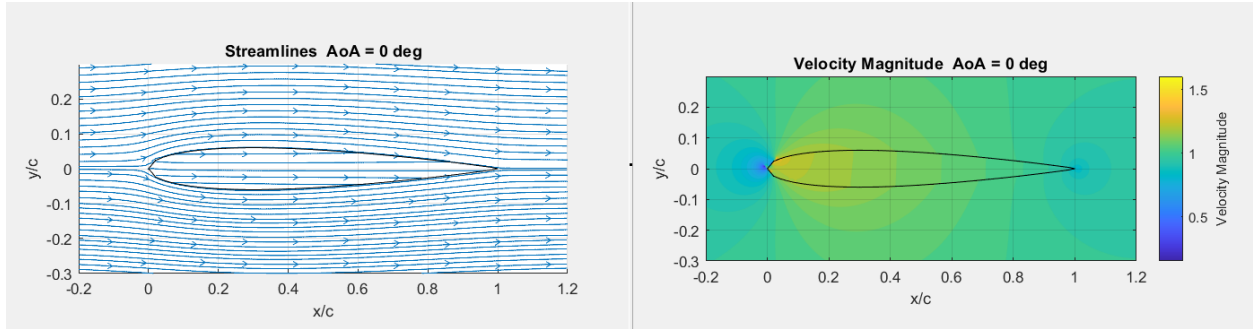


Figure 3: Streamlines (left) and velocity magnitude contours (right) for a NACA 0012 airfoil at $\alpha = 0^\circ$ using 100 panels. The flow field is symmetric around the chord with a smooth streamline and equal velocity distribution on both the upper and lower surfaces. Stagnation is observed near the leading edge.

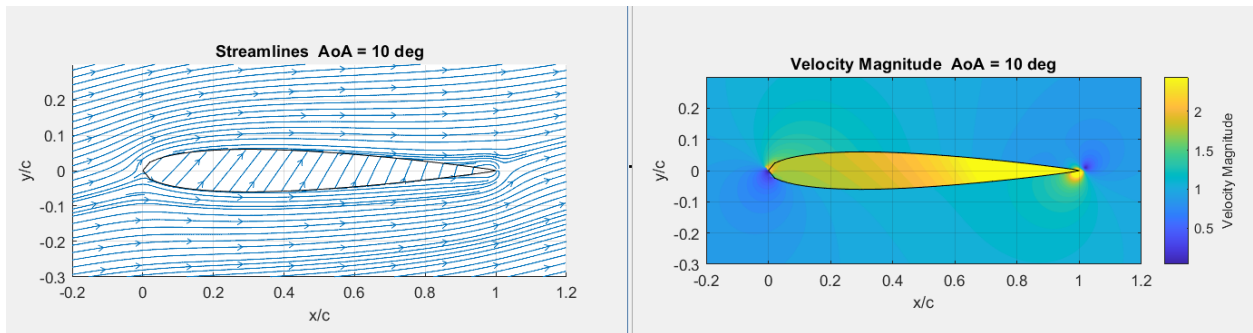


Figure 4: Streamlines (left) and velocity magnitude contours (right) for a NACA 0012 airfoil at $\alpha = 10^\circ$ using 100 panels. The flow field is asymmetric with increased velocity in the upper surface and the stagnation point shifted near the lower leading edge. The trailing edge is distorted due to the limitations of the source panel method.

The streamline and velocity magnitude contours are a visual representation of the flow field generated by the source panel method for a NACA 0012 airfoil at $\alpha = 0^\circ$ and $\alpha = 10^\circ$

At $\alpha = 0^\circ$, the flow field is symmetric about the chord, where the streamlines are smooth and divide the leading edge and recombine at the trailing edge, which indicates no lift and is accurate, as there is symmetric geometry and zero angle of attack. The velocity magnitude contours show low velocity nearing the leading edge, known as the stagnation point, followed by slight acceleration over both the upper and lower surfaces. The velocity distribution on both sides is symmetrical.

At $\alpha = 10^\circ$, the flow becomes asymmetric, with streamlines that deflect down as they pass over the airfoil. The stagnation point shifts along the lower surface near the leading edge, and the streamlines have more space over the upper surface, which shows higher flow speed. This

is also supported by the velocity magnitude contours, which have increased velocity along the upper surface and lower velocity along the lower surface.

The results, however, show that the source panel method has limitations, where near the trailing edge, the streamlines show unrealistic behavior with noticeable clusters and distortion, and the velocity contours exaggerated. This happens because the method does not include circulation or enforce the Kutta condition, which are necessary to produce an accurate lifting flow and smooth trailing edge. As a result, the flow is not realistic.

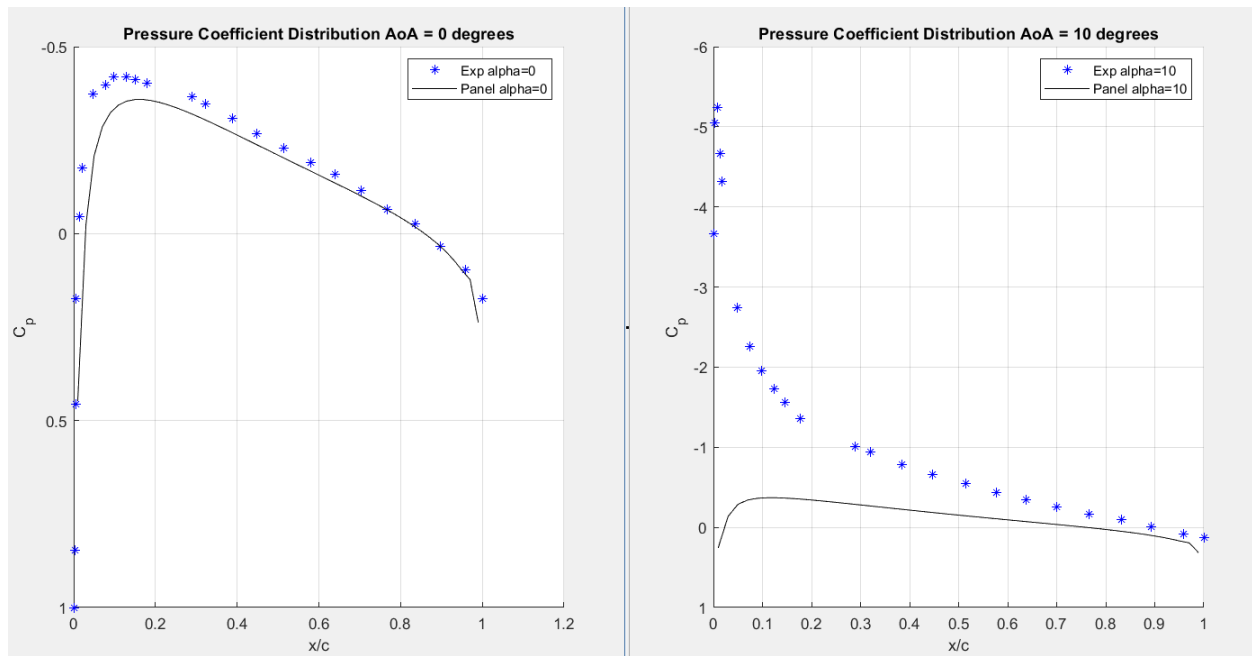


Figure 6: Pressure coefficient distribution for a NACA 0012 airfoil at $\alpha = 0^\circ$ and $\alpha = 10^\circ$, compared with experimental data. With no angle of attack, the panel method shows a general trend of the pressure distribution with slight discrepancies near the leading edge and trailing edge, while with slight angle of attack, the method underpredicts suction and fails to capture the right pressure distribution along the chord.

At $\alpha = 0^\circ$, the calculated pressure distribution shows consistent with the experimental data over most of the chord. The shape of the curve is shown as well as the drop in pressure coefficient near the leading edge and gradual recovery towards the trailing edge. However, differences are shown near the leading edge where the panel method suggests there to be more variation than the experimental data. This could be due to boundary-layer effects that smooth out the pressure in real time. Near the trailing edge, the pressure coefficient is slightly different from the experimental data, with a steeper decline and less smooth recovery, which could be due to the source panel method not accounting for the Kutta condition. Real flows have smooth flow at the trailing edge whereas here, the lack of Kutta condition prevents that.

At $\alpha = 10^\circ$, there are significant discrepancies. The experimental data shows suction near the leading edge with very low coefficient of pressure values, which suggests acceleration over the upper surface. This contrasts with the panel method which does not predict this suction, which produces noticeably smaller variation in the coefficient of pressure. This shows a major limitation of the source panel method; no circulation. Through the rest of the chord, the pressure distribution does not match the shape of the experimental data. The panel method predicts slight change in the coefficient of pressure, where the experimental data shows noticeably stronger pressure gradient and more recovery. This continues at the trailing edge where the source panel method lacks the smooth behavior that is seen in the experimental data due to the lack of a Kutta condition.

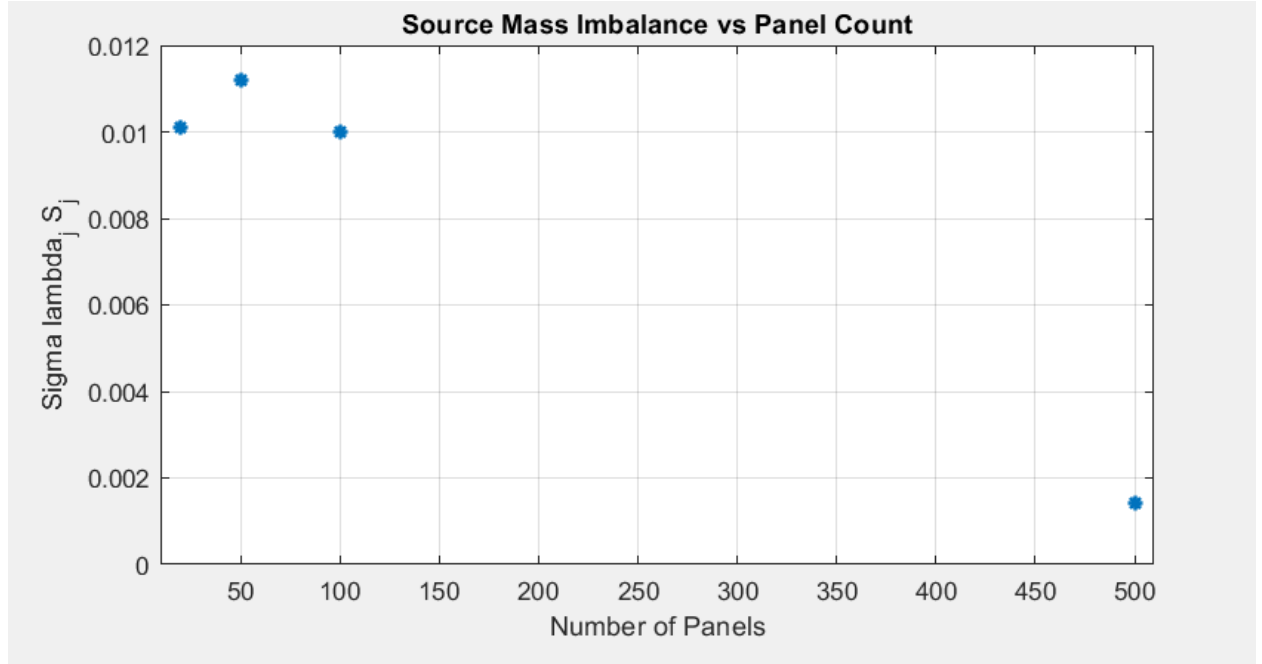


Figure 7: Source mass imbalance as a function of panel count for a NACA 0012 airfoil at $\alpha = 0^\circ$. The imbalance decreases with a panel increase, which indicates less error with more panels.

As described in the technical approach, the sum of all the source and sink strengths must equal to zero:

$$\sum_{j=1}^n \lambda_j S_j = 0$$

Any deviation from zero is the numerical error in the solution. The airfoil was evaluated at $\alpha = 0^\circ$ for varying panel counts.

Table 1: Computed source mass imbalance values

Number of Panels	$\sum_{j=1}^n \lambda_j S_j$
20	0.0100978
50	0.0111812
100	0.00999663
500	0.00140422

Total source mass imbalance decreases as the number of panels increase, showing continuingly less error. For lower panel counts, the imbalance remains in the order of 10^{-2} , which is significantly higher error than higher panel counts. At 500 panels, the imbalance is reduced noticeably, showing noticeably improved accuracy. This confirms that a higher number of panels is a better representation of the airfoil.

The trend between 20, 50, and 100 panels, where the imbalance increases before decreasing again is likely due to error accumulation but because the airfoil has noticeable reduction at 500 panels, this indicates that the method still converges despite variation.

Summary & Conclusions

The source panel method was successfully used to simulate flow over a NACA 0012 airfoil and determine its performance across different flow conditions. The method produced reasonable flow fields when there was symmetry.

At $\alpha = 0^\circ$, the results showed agreement with expectation, where the source strength distribution was symmetric, the streamlines were smooth and with even distribution, and the pressure coefficient followed the general trend of experimental data. There were minor discrepancies near the leading and trailing edges that were likely due to the lack of boundary layer and inviscid effects. At $\alpha = 10^\circ$, the method showed asymmetry in the flow including differences in velocity and source strength between the upper and lower surfaces. However the inaccuracies were observed in the pressure coefficient distribution, notably in the inability to predict the leading edge suction. Additionally, there was unrealistic behavior at the trailing edge that appeared in both the source strength distribution and flow field visualizations. These were likely due to the lack of circulation and Kutta condition within this method. The mass imbalance confirmed that the numerical solution increases in accuracy with the number of panels. While small variations were observed, the trend showed clear reduction in error.

In terms of computational performance, this method is relatively efficient for moderate panel counts and is an adequate solution for basic flow analysis. However, the increased number of panels improved the accuracy at the cost of noticeable computational time, especially when evaluating the velocity contour and streamlines.

The source panel method is useful for understanding how flow behavior works, but is limited in its accuracy for lifting flows. It would require including vortex panels as well as enforcing the Kutta condition to improve the methods ability for realistic simulation.